

TAX EVASION AND HEURISTICS

A research note

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The paper examines the heuristics which taxpayers use in making tax evasion decisions. The following propositions are tested in an experiment using students. First, the taxpayer's own level of evasion will be positively related to what he perceives as the levels of evasion by others. Second, taxpayers who have been audited are more likely to assess the probability of audit as higher and therefore decrease their levels of evasion. The experimental results fail to support the first proposition but do support the second suggesting the presence of an 'availability' effect.

1. Introduction

In this study, an attempt is made to examine the heuristics or 'rules of thumb' which taxpayers use in making tax evasion decisions, taking an experimental approach. A heuristic has been defined as a 'process for solving a problem which may aid in the solution but offers no guarantee of doing so' [Taylor (1965)]. The obvious shortcoming of an experimental approach lies in the artificiality of the laboratory setting which may make it difficult to generalize to tax evasion in the real world. However, this approach clearly provides more control than do other methods of analysis. Social psychologists, such as Thibaut et al. (1974), have made use of games and simulations in studying compliance with rules. Furthermore, the experimental approach to studying tax evasion has been taken by Friedland et al. (1978), Spicer and Becker (1980) and by Spicer and Thomas (1982). These experiments suggest that tax evasion is higher among those who feel disadvantaged by tax inequities and who perceive the probabilities of being audited as low. The fact that these results have been confirmed using survey research [Spicer and Lundstedt (1976), Song and Yarbrough (1978), Mason and Calvin (1978)] suggests that the experimental approach provides a useful, although obviously not the only, means of studying tax evasion.

2. Theoretical framework

According to economic models of tax evasion, such as developed by Allingham and Sandmo (1972) and Srinivasan (1973), tax evasion is a special

form of gambling; gambling for extra income in light of the likelihood of detection and penalties imposed on detected tax evaders. However, it is questionable whether taxpayers make the burdensome calculations necessary to determine an optimal level of tax evasion or whether they have the precise information necessary for such calculations. Heiner (1983) has argued that when individuals face a high degree of uncertainty they tend to fall back on rules or conventions in making decisions. This is because greater uncertainty will both reduce the chance of recognizing the right situation to select a particular action, and increase the chance of not recognizing the wrong situation for selecting it. Clearly, the rules which individuals use in place of optimization procedures are of interest.

One possibility is that the perceived behavior of others provides a guide for taxpayers. Certainly, a number of studies have suggested that tax evasion by an individual depends upon his perception of the tax evasion of others [Vogel (1974), Spicer and Lundstedt (1976), Song and Yarbrough (1978)]. Research by Kahneman, Slovic and Tversky (1982) suggests other 'rules of thumb' or heuristics which taxpayers may use. For example, they argue that individuals will tend to assess the probability of an event by the ease of recalling instances of similar events in the past. This is called the 'availability' heuristic.

The foregoing would suggest the following propositions. First, the taxpayer's own level of evasion will be positively related to what he perceives as the levels of evasion by others. Second, taxpayers who have been audited are likely to assess the probability as higher and therefore decrease their levels of evasion. Neither of these propositions would normally be derived from a model of tax evasion based on optimization procedures and it is these propositions that we sought to examine in the experiment.

3. Methodology

Thirty-six University of Colorado undergraduate students were recruited from psychology classes as participants for the experiment. They ranged in age from 17 to 50 years with a median age of 25. The procedures followed were similar to those used by Friedland, Maital and Rutenberg (1978). The participants were told that the research would consist of a tax game, consisting of ten pay periods (months). In each period they would be asked to decide how much of a given income they wish to report and then compute the taxes due on that amount. Audits would be conducted on randomly selected participants and fines imposed where taxes paid were less than taxes due on the monthly income. Participants were told that if they were audited, they would pay a fine equal to ten times the amount of tax evaded. At the end of the ten pay periods, a small money prize would be distributed to each participant. Each was told that the amount of his/her prize would be based

on total income minus taxes and fines paid. Finally, they were told that the average money prize would be \$8.00, but that their prize might be higher or lower.

Each participant was then given a folder containing tax forms, a tax table and a note of additional information. The note repeated and clarified information already received regarding audits and fines. However, 12 participants were told that in a previous game participants paid only 10 percent of taxes due. Another 12 were told that in a previous game participants paid 50 percent of taxes due. The rest were told that participants paid 90 percent of taxes due. This was in fact a deception since there had been no previous game.

After participants had finished reading the sheet of additional information, the experiment began. The 'monthly' cycle of income announcements, tax reporting and audits was repeated ten times. At the end of the game participants were asked to complete a short questionnaire in which demographic information was solicited. Upon completion of the questionnaire, prizes were computed. The experiment lasted approximately one hour.

4. Discussion of results

The amount of taxes evaded during the entire game was regressed on the level of evasion suggested in the 'previous game' (the game referred to in the note of information) and gender. As table 1 indicates, the amount of taxes evaded was not affected significantly by the information provided regarding evasion in the 'previous game'. Gender, however, did have a significant effect: men tended to evade more taxes than women. This is consistent with the findings of Spicer and Becker (1980).

Taxpayers, then, did not appear to use the behavior of others as a guide to their own evasion decisions. In order to examine the other proposition, the following analysis was performed. The level of evasion in the final (tenth)

Table 1
Taxes evaded with evasion in 'previous game' and gender (*t*-values in parentheses).

	Standardized regression coefficients
Evasion in 'previous game'	0.131 (0.90)
Gender ^a	0.535 ^b (3.68)
$R^2 = 0.303^b$	

^aFemale = 1.

Male = 2.

^bSignificant at 99 percent confidence level.

Table 2
Taxes evaded in the last round by taxes evaded in the first round
and the number of prior audits (*t*-values in parentheses).

	Standardized regression coefficients
Taxes evaded in first round	0.591 ^a (4.05)
Number of prior audits	-0.361 ^b (2.48)
$R^2 = 0.356^a$	

^aSignificant at a 99 percent confidence level.

^bSignificant at a 98 percent confidence level.

round of the game was regressed on the level of evasion in the first round and the number of audits in the first nine rounds. As one would expect, the amount of taxes evaded in the final round was positively and significantly related to tax evasion in the first round (table 2). Moreover, the number of audits had a significant and negative effect on tax evasion in the last round. This latter finding suggests the presence of the 'availability' effect and is consistent with our proposition.

5. Implications

The experiment failed to demonstrate that tax evasion behavior is significantly affected by information provided regarding the behavior of others. It may be that the relationship between tax evasion and the perceived behavior of others, which has been observed in survey research, depends upon a 'social stigma' attached to tax evasion behavior. The fact that these stigmatization effects were probably not present in the experimental setting may explain the lack of finding here. However, the results would seem to suggest that taxpayers do not engage in optimizing strategies in making tax evasion decisions. Instead, they apparently use heuristics or 'rules of thumb'. In particular, the experience of being audited appears to lower levels of tax evasion even where the likelihood of an audit is completely random. Our analysis suggests, then, that random audits may lead to significantly higher levels of compliance among those audited, a result not predicted by conventional economic models. Further research into the rules or conventions underlying tax evasion decisions would seem warranted.

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